



Cambrian College

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality

Course Outline

Course Title	Software Testing Fundamentals				
Course Code:	STY1000	Credit Value:	4	Credit Hours:	56
Programs:	BAGC I.T. Business Analysis BTHP I.T. Business Analysis - Hanson				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

General Education Course: ☐

Eligible for PLAR: ☒

COURSE DESCRIPTION

In this course, students will learn the fundamentals of software testing. The course content extends existing software skills and follows the ISTQB-BCS Certified Tester Foundation Level Curriculum. The aim of the course is to prepare students for roles such as testers, test analysts, test engineers, test consultants, test managers, user acceptance testers and software developers.

Date: June 19, 2020

Prepared By: Nathan Abourbih

Approved by: 

BRIAN LOBBAN
Dean, School Of Business And Information Technology,
Creative Arts, Design, Music And Hospitality

Effective: Fall 2020, Winter 2021, Spring 2021

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL	
This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:	
Program(s)	Vocational Learning Outcomes
I.T. Business Analysis	College Standards 1. Document business processes using effective and established research and documentation methods and business improvement procedures. 2. Analyze the information collected during the requirements elicitation to identify gaps and define the capabilities of an acceptable solution using appropriate methods, tools, techniques and documentation. 3. Develop test strategies to ensure that business requirements and stakeholder objectives for changes to a business system are met.
I.T. Business Analysis - Hanson	Ministry Standards 1. Document business processes using effective and established research and documentation methods and business improvement procedures. 2. Analyze the information collected during the requirements elicitation to identify gaps and define the capabilities of an acceptable solution using appropriate methods, tools, techniques and documentation. 3. Develop test strategies to ensure that business requirements and stakeholder objectives for changes to a business system are met.

COURSE CURRICULUM**Topics/Concepts Covered in This Course**

- Fundamentals of Testing
- Life Cycles
- Static Testing
- Test Techniques
- Test Management
- Tool Support for Testing

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	Objectives
1. Define the fundamentals of software testing and demonstrate appropriate use of terminology related to the development and testing of software.	1.1 Summarize the fundamentals of software testing and give examples of industry-recognized certifications and their value. 1.2 List common reasons that software fails and explain why testing is an essential part of preventing issues.

Learning Outcome	Objectives
	1.3 Describe what software testing is and explain what it does to improve a software project's outcome. 1.4 Discuss the general principals of testing and the importance of keeping software under control. 1.5 Examine the effects that psychology has on software testing. 1.6 Identify the most common difficulties involved with software testing and describe how they can be mitigated.
2. Examine the role of software testing as it relates to the Software Development Life Cycle.	2.1 Define and analyze the various components, models, and methodologies related to software development and testing. 2.2 List the various levels of testing and describe how and where they fit into the software development process. 2.3 Identify the most common types of software testing performed today and define their characteristics. 2.4 Explain the maintenance testing process as it relates to the enhancement or migration of software that has already been deployed.
3. Use the technique of static testing to examine a piece of software for defects.	3.1 Define static testing and discuss the various aspects of using this technique. 3.2 Discuss the advantages and disadvantages of the static testing technique and compare and contrast it with the dynamic testing technique. 3.3 Explain the review process of static testing and how it is used to find potential defects in a piece of software. 3.4 Identify the most common static testing tools used today and compare the benefits of each. 3.5 Use static testing tools and techniques to identify a variety of different types of defects located in a piece of software.
4. Evaluate a given scenario and select the most appropriate software testing technique.	4.1 Define the test development process. 4.2 Explain the test case design technique and discuss the importance of test coverage. 4.3 Describe specification-based (black-box) techniques. 4.4 Describe structure-based (white-box) techniques. 4.5 Describe experience-based techniques. 4.6 Choose the best test technique for a given scenario and justify the choice.
5. Show what it takes to manage the entire software testing process.	5.1 List and describe the various phases of the test management process. 5.2 Explain the process of identifying risk factors and finding possible solutions.

Learning Outcome	Objectives
	5.3 Identify and describe the components of test planning through the definition of objectives, scope, strategy, scheduling, resources, and communication. 5.4 Discuss the characteristics of the perfect test team. 5.5 Describe the importance of feedback and control during testing. 5.6 Explain configuration management as it relates to testing. 5.7 Illustrate how problems that occur during the software development and testing process can be managed. 5.8 Use software tools that are commonly used in test management today.
6. Demonstrate the use of software tools to perform a test.	6.1 List the types of test tools and describe their function. 6.2 Describe the potential benefits and risks of using test tools and discuss their effective use. 6.3 Explain how to introduce a tool into an organization. 6.4 Identify the most common test tools used today and compare the purpose and benefits of each. 6.5 Use a software test tool to examine a piece of software for potential defects.

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.
- respond to written, spoken, or visual messages in a manner that ensures effective communication.

Numeracy

- execute mathematical operations accurately.

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.
- use a variety of thinking skills to anticipate and solve problems.

Information Management

- locate, select, organize, and document information using appropriate technology and info systems.
- analyze, evaluate, and apply relevant information from a variety of sources.

Interpersonal

- not applicable

Personal

- manage the use of time and other resources to complete projects.
- take responsibility for one's own actions, decisions, and consequences.

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Online: Course is fully delivered through asynchronous teaching.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.
- HyFlex: Course includes both synchronous and asynchronous learning and the student can move between both components seamlessly.

Learning Activities

- Lectures
- Class Discussions
- Labs
- Group Work
- Research
- Presentations
- Scenario Testing
- In-Class Exercises
- Case Studies

Resources Required

Books

Geoff Thompson, *Software Testing: An ISTQB-BCS Certified Tester Foundation guide*, 4th, BCS, the Chartered Institute for IT
ISBN: 978-1780174921

Evaluation Plan

Grading Scheme

A	80% - 100%
B	70% - 79%
C	60% - 69%
D	50% - 59%
F	0% - 49%

Evaluation Method	Value (%)
Assignments	40%
Applied Activities(G)	40%
Quizzes/Test/Exam	20%

ADDITIONAL INFORMATION

College

Academic Policies

Students must adhere to all academic policies. These are available on myCambrian as well as the College website at <https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/>.

Prior Learning Assessment and Recognition (PLAR)

Students may contact the Pathways office at pathways@cambriancollege.ca

Transfer Credit

Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Accessibility Services

Cambrian is committed to creating an inclusive learning environment where we support our students' success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@cambriancollege.ca

Testing Practices at Cambrian

Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Wabnode Centre for Indigenous Services

Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling or academic tutoring. Our team will also be able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode for cultural, sports and community activities. Students can also contact Wabnode at wabnode@cambriancollege.ca

Copyright

Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: <http://cambriancollege.libguides.com/copyrightinfo> or by contacting the Library for questions related to copyright: library@cambriancollege.ca

Out-of-Class Assistance

The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Audio/Visual Capture

Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Equity, Diversity and Inclusivity

Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: <https://cambriancollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/>

Tests and Evaluations

Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Attendance

Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Modifications to the Course Evaluation Plan

Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality Regulations**Lab Attendance**

Lab attendance is mandatory since lab grades include an evaluation of lab reports and lab performance. Lab performance can only be judged in person, requiring that students attend all scheduled labs.